

# Lakshya Verma

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## Summary

CS undergraduate (AI/ML, graduating 2027) who builds and ships **end-to-end data and AI products** — from LLM agents and RAG pipelines to forecasting models and interactive dashboards. Hands-on through two internships (AI and finance/operations) and a portfolio of deployed projects spanning text-to-SQL agents, predictive ML, and business intelligence. Comfortable owning a problem from messy raw data to a working, decision-ready result, in **Python** and **SQL**.

## Education

**Manipal Institute of Technology, Bengaluru**

Aug 2023 – May 2027

*B.Tech, Computer Science & Engineering (AI/ML) | Minor: Data Analytics*

**Relevant Coursework:** Machine Learning, NLP, Database Management Systems, Business Intelligence, Data Analytics, Statistics.

## Technical Skills

- **AI & LLMs:** LLM prompting & fine-tuning, RAG pipelines, text-to-SQL, AI agents / agentic workflows, NLP (sentiment, theme extraction); Google Gemini, OpenAI, Anthropic APIs
- **Data & Programming:** Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (PostgreSQL, MySQL, Microsoft SQL Server), data modeling, REST APIs, Git
- **Analytics & BI:** Power BI, DAX, Plotly Dash, Streamlit, Excel (Pivot Tables, Power Query); EDA, forecasting, data storytelling
- **ML & Automation:** Predictive modeling (classification, clustering, regression), statistical analysis, n8n workflow automation, automated reporting, data-quality assurance

## Experience

**1Stop.ai (Languify) — Artificial Intelligence Intern**

May 2025 – Jun 2025

*Remote*

- Restructured AI training pipelines with NLP-based preprocessing and data cleaning, cutting preprocessing time by 30% and accelerating model iteration cycles.
- Evaluated model performance with ML metrics and translated findings into actionable insights (+20% optimization efficiency); built automated narrative summaries for stakeholders.

**Nirmala Swarup Securities Pvt. Ltd. — Data & Operations Intern**

Dec 2025 – Feb 2026

*Agra, India*

- Built structured data models and workflows across Excel and Tally ERP, reducing data discrepancies by 35% and improving compliance accuracy.
- Designed interactive dashboards and SQL reports surfacing operational KPIs, cutting reporting time by 40% and enabling faster, data-backed decisions.

## Projects

**Natural Language to SQL Query Agent**

[GitHub](#)

*Google Gemini, n8n, PostgreSQL, LLMs, RAG*

- Built an AI agent using text-to-SQL and RAG to translate natural-language queries into optimized SQL, enabling self-serve data access for non-technical users and cutting query time by 60%.
- Iterated on LLM prompting with structured error analysis (+35% output precision) and generated automated narrative reports from query results.

**Tamil Nadu Elections Decoded — Electoral Analytics Pipeline**

[GitHub](#) | [Live](#)

*Python, Pandas, Plotly, Streamlit, python-pptx*

- Built an end-to-end pipeline ingesting 8,489 records across 234 constituencies, standardizing 13+ source variants and detecting flips; quantified a +12.06 pp turnout swing and a seat-volatility index.
- Auto-generated broadcast-ready executive decks (PowerPoint + PDF) and a logistic-regression swing-seat classifier in pure NumPy/Pandas.

**Trader Sentiment Analysis — Predictive ML Model**

[GitHub](#) | [Live](#)

*Python, Scikit-learn, K-Means, Random Forest*

- Merged ~211K trade fills with the Bitcoin Fear & Greed Index, segmented traders via K-Means, and trained a Random Forest to predict next-day PnL direction across market-sentiment regimes.

**Automated Data Governor — Data Quality & Integrity**

[GitHub](#)

*Python, PostgreSQL, n8n, AI Agents*

- Shipped an AI-driven tool that audits PostgreSQL databases for integrity issues and triggers automated alerts and SQL quality reports, cutting manual audit time by 50% and resolution time by 45%.